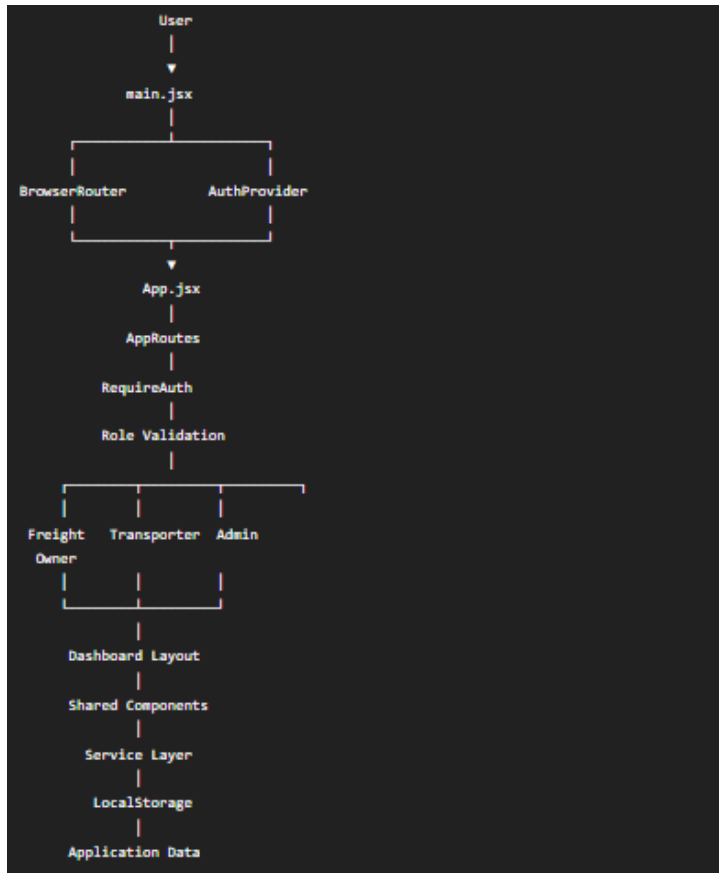


Front-End Architecture Overview

The Truck Asset Matchmaking Platform (TAMP) is built as a modular **React + Vite** application using **React Router**, **Context API**, and **LocalStorage**. Its architecture is designed for scalability, maintainability, and clear separation of concerns.

Architecture Flow



Core Design

- **React Router** manages navigation and protected routes.
- **Context API** provides global authentication and user state.
- **Role-based routing** ensures users access only authorised pages.
- **Dashboard Layout** offers a consistent interface across all roles.
- **Services** separate business logic from UI components.
- **LocalStorage** simulates backend persistence for users and application data.

Key Principles

- **Separation of Concerns** – UI, routing, authentication, and data logic are independent.
- **Role-Based Structure** – Features are organised by Freight Owner, Transporter, and Administrator.
- **Reusable Components** – Shared layouts and UI components reduce duplication.
- **Centralised Routing** – All application routes are managed from a single configuration.

Summary

The application follows a role-based, component-driven architecture where routing controls access, Context API manages authentication, services handle data operations, and reusable components provide a consistent user experience. This design makes the application modular, maintainable, and easy to extend.

